VP150 Final Big RC Part 2

Outline

Non-inertial FOR

Exercise

Translation Non-inertial FOR

Definition

$$a = a' + a_0$$

$$F = ma = m(a' + a_0)$$

$$F + F_{inertial} = ma' = ma + (-ma_0)$$

Rotation Non-inertial FOR

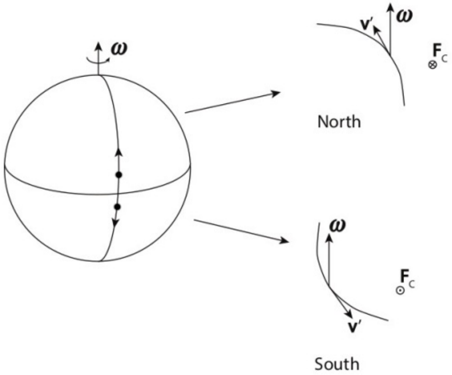
Definition

$$a = a_0 + a' + 2\omega \times v' + \frac{d\omega}{dt} \times r' + \omega \times (\omega \times r')$$

$$ma' = ma - ma_0 - m\frac{d\omega}{dt} \times r' - 2m(\omega \times v') - m\omega \times (\omega \times r')$$

$$ma' = F + F_{inertial}$$

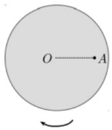
Coriolis Force in Nature



Outline

Exercise 1

(*5 points*) A platform rotates clockwise about the axis perpendicular to the page and through the platform's center O, as shown in the top-view figure. The angular speed of rotation is constant. Suppose that you are standing at point A, throwing a ball towards the center of the platform O, along the radial direction AO.

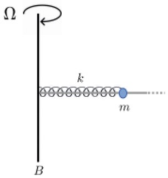


(a) An observer, standing on the platform at O, will see the object being deflected to the left or to the right with respect to the radial direction AO (as he looks from O)?

(b) How will an immobile observer outside of the platform explain this deflection?

Exercise 2

(*7 points*) [*Note.* There is no gravitational force in this question.] A linear spring with spring constant k and unstretched/uncompressed length is x_0 is coiled around a light long horizontal rod (see the figure). The rod is fixed to a vertical axle B, which spins always with the same angular speed $\Omega = \sqrt{k/m}$. One end of the spring is fixed at the axle. At the other end of the spring there is bead with mass m that can slide along the rod, but the rod is rough so that the bead moves with friction.



What should the value of the coefficient of kinetic friction μ be, so that the bead slides away from the axle with constant speed v_0? Comment briefly on whether the scenario described in this problem is sustainable?

Reference

- Yinghui He, 22SU VP150 RC.
- Yufan Wu, 22SU VP150 RC.
- Mateusz Krzyzosiak, 2023SU VP150 Slides.